What Does Swiss German Cost a Language Model? Separating Dialect from Spelling in LLM Normalization

Erik Avtandilyan and Simone Martinecz and Kelvin Myers
University of Zurich
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https://github.com/Eroavt/gsw-eval

Abstract

Swiss German is spoken by about five million people but has no standard spelling, so every writer spells it their own way. Large language models (LLMs) understand it worse than Standard German and need more tokens for it. We ask how much of this cost comes from the dialect and how much from the spelling. We build GSW-EVAL, an open, reproducible benchmark that feeds the same 500 sentences to a model as Standard German, as careful Swiss German in eight dialects, and as texting-style and character-noised Swiss German. On SwissDial, the dialect costs Apertus 8B 20.1 chrF points and Llama 3.1 8B 24.4 points; texting spelling adds only 3.4 and 4.9. The Swiss model Apertus beats Llama by 7.1 points, and the gap grows under noisy spelling. Token counts tell the same story: Swiss German needs 16 to 30% more tokens than Standard German, while texting spelling changes the count by about 1%. Per-dialect difficulty follows how close each written dialect is to Standard German ($r = 0.93$). **[TODO: one sentence on the fresh set and the human ratings once they are in.]**

1 Introduction

Written Swiss German is everywhere in Switzerland: in messages, social media and comments. It is also a textbook case of a low-resource variety inside a high-resource language: Standard German is one of the best-served languages in NLP, its dialects are not (Joshi et al., 2020). Two things make Swiss German hard for a language model. First, the *dialect*: the eight major dialects differ in vocabulary, sound and grammar. Second, the *spelling*: there is no orthography, so the same writer may spell a word differently from one message to the next.

Users pay for both, in quality and in money, since API prices are per token and non-standard text splits into more tokens (Petrov et al., 2023; Ahia et al., 2023). But the two causes need different fixes. A dialect problem calls for more dialect data; a spelling problem could be solved with normalization or with robust training. Existing work measures one or the other. We measure both on the same sentences.

Contributions.

1. GSW-EVAL, an open benchmark that runs any LLM on written Swiss German and splits the quality loss into a *dialect cost* and a *spelling cost*, with confidence intervals.
2. A comparison of a Swiss and a non-Swiss model of the same size, plus a commercial reference, showing that **[TODO: headline finding, final wording after the Claude run]**.
3. A token-tax analysis of nine tokenizers, showing that the extra tokens come from the dialect, not from the spelling.
4. A small fresh test set in which each native writer translates the same sentences twice, carefully and in texting style. **[TODO: size once collected]**

2 Related Work

[TODO: Kelvin: rewrite this section in your own words, 0.5 column.]

Swiss German NLP. Earlier work normalizes Swiss German transcripts to a common spelling with character-level machine translation (Scherrer and Ljubešić, 2016) on the ArchiMob corpus (Samardžić et al., 2016). SwissBERT (Vamvas et al., 2023) adapts an encoder to Swiss national languages, and Vamvas et al. (2024) adapt encoders to written Swiss German. SwissDial (Dogan-Schönberger et al., 2021) provides the same sentences in eight dialects with Standard German references, which makes a controlled comparison across dialects possible. Collecting dialect data remains the bottleneck (Sun et al., 2025).

Dialect and noise robustness. DialectBench (Faisal et al., 2024) shows large gaps between standard and non-standard varieties across many tasks. Synthetic character noise breaks neural translation (Belinkov and Bisk, 2018), and injecting such noise during training helps transfer between closely related languages (Aepli and Sennrich, 2022). We use the same noise for evaluation and compare it with rule-based texting spelling.

Tokenization cost. The same content needs very different numbers of tokens in different languages (Rust et al., 2021; Petrov et al., 2023), which makes commercial models more expensive for some users (Ahia et al., 2023). We apply this lens *within* one language: Standard German versus its dialects.

3 Data

SwissDial. SwissDial v1.1 (Dogan-Schönberger et al., 2021) contains 11,213 Standard German sentences with transcriptions in up to eight dialects (AG, BE, BS, GR, LU, SG, VS, ZH), one writer per dialect. We keep the 2,529 sentences available in all eight dialects and drop 232 marked as code-switching, leaving 2,297. We draw a fixed split of 50 development and 500 test sentences (seed 13), giving 4,000 Swiss German test inputs per condition. The Standard German sentence is always the reference.

Fresh set. **[TODO: Simone: who wrote it (number of writers, dialects), how long it took.]** Each writer translated the same 30 Standard German sentences (20 everyday chat, 10 news-like) twice: once carefully and once "the way you would text". Because the same person writes both versions, the difference between them is spelling only. The set was written after the models were trained, so it cannot be in their training data. It is released with the code (CC BY 4.0).

4 Method

Conditions. Every test sentence is given to each model in four forms: (1) *Standard German input*: the reference itself; (2) *clean*: the SwissDial transcription; (3) *texting*: the clean text rewritten by 24 texting rules (lower case, dropped punctuation, collapsed double vowels, short forms such as *vlt* for *vilicht*), each firing with a probability, **[TODO: Simone: confirm the rules are realistic]**; and (4) *character noise*: 10% of words get one random letter insertion, deletion or substitution, following Aepli and Sennrich (2022). All noise is seeded per sentence.

Baselines. The *identity* baseline copies the Swiss German input unchanged; it is the lower bound every model must beat. The *Standard German input* condition is the upper bound: the best a model can do when there is no dialect and no spelling problem.

Dialect and spelling cost. We define the *dialect cost* as chrF on Standard German input minus chrF on clean Swiss German, and the *spelling cost* as chrF on clean minus chrF on texting input. Both are computed on the same sentences, so content is held constant.

Models. Apertus 8B Instruct (Hernández-Cano et al., 2025), trained with a strong Swiss and multilingual focus, and Llama 3.1 8B Instruct (Grattafiori et al., 2024), the same size without a Swiss focus, both run locally with 8-bit weights and greedy decoding via MLX. Claude Sonnet 5.5 serves as a commercial reference. **[TODO: Claude results.]** All models receive the same German prompt asking for the Standard German sentence only.

Metric and statistics. We score with chrF (Popović, 2015) as implemented in sacreBLEU (Post, 2018); character n-grams suit a task where most errors are partial word forms. All intervals are 95% bootstrap intervals over sentences (Koehn, 2004); model differences use a paired bootstrap, and the costs use a cluster bootstrap that resamples whole sentences with all their dialects. We remove answer preambles such as "Der Satz lautet auf Standarddeutsch:", which 4.5% of 8B outputs contain.

Token tax. For a tokenizer, the token tax is the number of tokens of the Swiss German sentences divided by the tokens of the same sentences in Standard German, summed over all 2,297 parallel sentences and eight dialects. Because the texting rules drop punctuation and punctuation marks are tokens, we compare all conditions with punctuation removed.

Human evaluation. **[TODO: Kelvin + Simone: two native speakers rated 96 outputs (48 inputs x 2 models, blind, random order) for adequacy on a 1–5 scale; report kappa here.]**

5 Results

The dialect costs far more than the spelling. Table 1 shows that moving from Standard German to clean Swiss German costs Apertus 20.1 and Llama

Model	Std. German	Clean	Texting	Char. noise	Dialect cost	Spelling cost
Identity (lower bound)	100.0 ±0.0	46.9 ±0.4	40.0 ±0.3	45.3 ±0.4	53.1 ±1.0	6.8 ±0.4
Apertus 8B	94.0 ±1.0	74.0 ±0.5	70.6 ±0.6	71.2 ±0.5	20.0 ±1.2	3.4 ±0.5
Llama 3.1 8B	91.5 ±1.0	66.9 ±0.6	61.9 ±0.7	63.9 ±0.6	24.7 ±1.3	4.9 ±0.7

Table 1: chrF on the 500 test sentences (±: half-width of the 95% bootstrap interval). Swiss German columns pool 4,000 inputs over eight dialects. Dialect cost = Standard German input minus clean; spelling cost = clean minus texting.

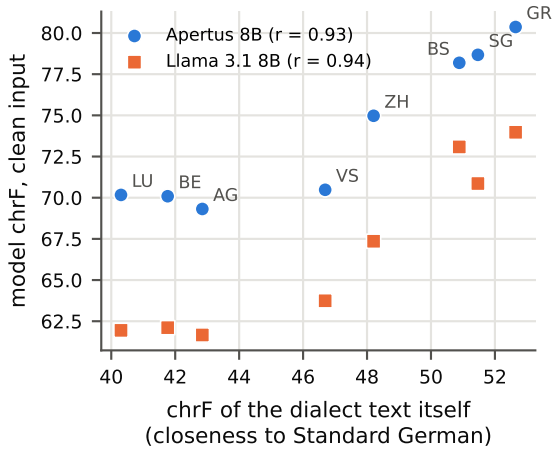


Figure 1: Per-dialect chrF of each model (clean input) against the chrF of the untranslated dialect text, a measure of how close its spelling is to Standard German.

Tokenizer	Tok./word	Tax	Range	Texting
Apertus 8B	1.64	1.30	1.23–1.40	+0.3%
Llama 3.1 8B	1.93	1.17	1.12–1.25	-1.3%
Gemma 3	1.62	1.25	1.20–1.35	+0.0%
Qwen 3	1.92	1.18	1.12–1.26	-1.0%
Mistral 7B v0.3	2.10	1.16	1.11–1.23	-0.7%
OpenAI o200k	1.60	1.27	1.23–1.37	-0.4%
XLM-R	1.63	1.30	1.25–1.39	+0.8%
SwissBERT	1.43	1.59	1.48–1.70	+7.1%

Table 2: Token tax per tokenizer. Tok./word: tokens per word of Standard German. Tax: tokens for Swiss German divided by tokens for the same Standard German sentences (punctuation removed, eight dialects pooled). Range: cheapest to most expensive dialect. Texting: change of the tax caused by texting spelling.

24.4 chrF points. Texting spelling costs a further 3.4 and 4.9 points, and character noise 2.8 and 3.0. Both models stay far above the identity baseline (46.7), so they do translate rather than copy.

The Swiss model is better, and more robust. Apertus beats Llama in every condition (paired bootstrap, $p < 0.001$): by 2.8 points on Standard German input, 7.1 on clean dialect and 8.6 on texting input. The advantage is largest where the input is least standard, which suggests that exposure to Swiss text helps most where spelling varies.

Even Standard German is not copied. Neither model reaches 100 on Standard German input: Apertus returns the sentence unchanged 64% of the time, Llama only 45%; the rest are paraphrases such as "Jemand stört mich hier bei meiner Arbeit" for "Irgendjemand pfuscht mir hier ins Handwerk". This is why the upper bound matters: part of every model's gap to 100 is paraphrase that chrF counts as error, not misunderstanding. **[TODO: the human ratings tell how big this part is.]**

Hard dialects are the ones written farthest from Standard German. Scores range from 69.0 (AG) to 80.0 (GR) for Apertus, and the ranking is almost the same for Llama (Figure 1). Both correlate strongly with the identity score of each dialect ($r = 0.93$ and 0.94): models do worst where the written form is farthest from Standard German. Since SwissDial has one writer per dialect, this mixes the dialect with the writer's personal spelling. The fresh set separates the two. **[TODO: fresh-set result.]**

Token tax comes from the dialect, not the spelling. Swiss German needs 16% (Mistral) to 30% (Apertus, XLM-R) more tokens than the same content in Standard German (Table 2). Texting spelling changes this by about 1% for every LLM tokenizer: it creates rarer word forms but also shortens words, and the two effects cancel. Bernese is the most expensive dialect and Zurich German the cheapest for all LLM tokenizers. Apertus pays the highest *relative* tax only because its tokenizer is very efficient on Standard German (1.64 tokens per word against 1.93 for Llama); in absolute terms it needs fewer tokens for Swiss German than Llama. **[TODO: price per 1,000 sentences for Claude once the key is set.]**

Robustness. **[TODO: prompt check (3 prompts) and full precision vs 8-bit, from the MacBook Pro runs: does the ranking change?]**

Human ratings. **[TODO: kappa, Spearman chrF vs rating, and whether chrF picks the same better model as the raters.]**

Fresh set. **[TODO: real vs synthetic texting cost, per writer.]**

6 Discussion and Limitations

[TODO: Kelvin + Erik: write after all results are in. Points to cover:] (1) SwissDial transcriptions are free translations by one writer per dialect, so dialect and writer are confounded; the fresh set addresses this only for **[TODO: N]** writers. (2) The texting rules are hand-written and applied with fixed probabilities; real texting is richer (abbreviations, emoji, code-switching). (3) chrF punishes valid paraphrases; the human ratings estimate how much. (4) We test one prompt per model in the main run and 8-bit weights; the robustness checks cover both on a subset. (5) SwissDial has been public since 2021 and may be in the models' training data; the fresh set cannot be.

7 Conclusion

[TODO: 3–4 sentences: answer the question in the title with the two numbers.]

Code and Data

Code, outputs of every model, and the fresh set: `https://github.com/Eroavt/gsw-eval`. Every number in this report can be reproduced with one command, `scripts/reproduce.sh`, which needs neither a GPU nor an API key because all model outputs are cached.

Use of AI Assistants

We used an AI coding assistant (Claude Code) to write and test the code and to draft parts of this report. All design decisions, the data collection and the final text are our own, and we checked every number against the code's output. **[TODO: adapt to UZH rules.]**

References

Noëmi Aepli and Rico Sennrich. 2022. Improving zero-shot cross-lingual transfer between closely related languages by injecting character-level noise. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 4074–4083.

Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David Mortensen, Noah Smith, and Yulia Tsvetkov. 2023. Do all languages cost the same? tokenization in the era of commercial language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9904–9923.

Yonatan Belinkov and Yonatan Bisk. 2018. Synthetic and natural noise both break neural machine translation. In *International Conference on Learning Representations (ICLR)*.

Pelin Dogan-Schönberger, Julian Mäder, and Thomas Hofmann. 2021. SwissDial: Parallel multidialectal corpus of spoken Swiss German. *arXiv preprint arXiv:2103.11401*.

Fahim Faisal, Orevaoghene Ahia, Aarohi Srivastava, Kabir Ahuja, David Chiang, Yulia Tsvetkov, and Antonios Anastasopoulos. 2024. DIALECTBENCH: An NLP benchmark for dialects, varieties, and closely-related languages. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 14412–14454.

Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Alejandro Hernández-Cano et al. 2025. Apertus: Democratizing open and compliant LLMs for global language environments. *arXiv preprint arXiv:2509.14233*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293.

Philipp Koehn. 2004. Statistical significance tests for machine translation evaluation. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395.

Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. 2023. Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Maja Popović. 2015. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395.

Matt Post. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191.

Phillip Rust, Jonas Pfeiffer, Ivan Vulić, Sebastian Ruder, and Iryna Gurevych. 2021. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pages 3118–3135.

Tanja Samardžić, Yves Scherrer, and Elvira Glaser. 2016. ArchiMob – a corpus of spoken Swiss German. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*.

Yves Scherrer and Nikola Ljubešić. 2016. Automatic normalisation of the Swiss German ArchiMob corpus using character-level machine translation. In *Proceedings of the 13th Conference on Natural Language Processing (KONVENS 2016)*, pages 248–255.

J. Sun, R. Sevastjanova, S. Ahmadi, R. Sennrich, and M. El-Assady. 2025. Dia-Lingle: A gamified interface for dialectal data collection. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (System Demonstrations)*, pages 148–158.

Jannis Vamvas, Noëmi Aepli, and Rico Sennrich. 2024. Modular adaptation of multilingual encoders to written Swiss German dialect. In *Proceedings of the 1st Workshop on Modular and Open Multilingual NLP (MOOMIN 2024)*, pages 16–23.

Jannis Vamvas, Johannes Graën, and Rico Sennrich. 2023. SwissBERT: The multilingual language model for Switzerland. In *Proceedings of the 8th Swiss Text Analytics Conference (SwissText 2023)*, pages 54–69.